

Microeconomics Analysis: Decision Theory I

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2026

Reading List

- D. Kreps “Microeconomic Foundations: Choice and Competitive Markets”, Chapters 1-4 and 7.
- A. Mas-Colell, M. Whinston and J.Green, “Microeconomic Theory”, Chapters 1-3.

1 Preferences

2 Utility

3 Choice

4 Bundles and Time

Introduction

Microeconomics focuses on explaining and predicting choice.

An intuitive idea is that agents choose the most desirable alternative from a set of feasible ones:

- interpret this is more like “as if”
- we need to define what feasible set means for a specific context
- desirability is represented by preferences, which describe choice in various environments

Preferences

In most economic models, preferences are considered to be:

- **Ordinal:** Which options are better, without quantifying the preference. This simplifies the requirements on the preference system, yet creates a rich set of predictions and questions.
 - The indifference curve represents ordinal preferences: these curves can be drawn without specifying how “far” curves are from each other.
 - On the contrary, cardinal preferences represent the measure of difference between the options.
- **Stable:** Do not change from (choice) problem to problem. Provides good predictability.

Binary Relations: Definitions

Formally, we analyse Binary Relations

Binary Relation

Given a set of alternatives X , a binary relation is a description of pairs of elements in X that are *related*

- We denote $(x, y) \in B$ or simply xBy
- Thus, a binary relation is a subset of $X \times X$
- We often use $\succsim$, $\succ$, P when the binary relation aims to represent a preference
- Many helpful representations: graphs, matrices, etc.
- Do not forget that, in a binary relation, pairs are ordered.

Preferences as Binary Relations

Donors and Receptors

$X = \{A, B, AB, 0\}$ (red cell) blood types

The binary relation ... **can donate red cells to** ...

Majority Voting

$X = \{x, y, z\}$ are three political candidates forming a Condorcet Cycle.

The binary relation **a majority of voters prefers candidate ... to candidate ...**

Lexicographic Preferences on $X = \mathbb{R}_+^2$

The binary relation ... **is at least as good as** ... defined by:

A consumer wants more of each good and no trade-off exists between goods. More of good 1 unequivocally makes a bundle better and a (larger) amount of good 2 is only relevant when the amount of good 1 is the same. That is, $(x_1, x_2)L(y_1, y_2) \Leftrightarrow x_1 > y_1 \text{ or } (x_1 = y_1 \text{ and } x_2 \geq y_2)$.

Binary Relations: Reflexivity and Completeness

Binary relations that capture preference concepts must have a series of properties:

Definition (Reflexivity)

For every $x \in X$, xBx must be true

- Our three examples were reflexive
- The parental binary relation is not reflexive

Definition (Connectedness)

B is connected if for every $x, y \in X : x \neq y$, either xBy , yBx or both must be true

Definition (Completeness)

B is complete if or every $x, y \in X$, either xBy , yBx or both must be true

Proposition

- 1 If B is complete, then it is reflexive
- 2 If B is connected and reflexive, then it is complete

Reflexivity and Completeness: Examples

- Parent-child relation is not reflexive, hence not complete.
- Blood donor relation is reflexive, but not connected/complete:

① $O \rightarrow \{O, A, B, AB\}$

② $A \rightarrow \{A, AB\}$

③ $B \rightarrow \{B, AB\}$

④ $AB \rightarrow \{AB\}$

⑤ Hence, our binary relation is:

$$d = \{(O, O), (O, A), (O, B), (O, AB), (A, A), (A, AB), (B, B), (B, AB), (AB, AB)\}$$

and $(A, B), (B, A) \notin d$.

- The majority voting relation is complete.
- The lexicographic relationship is complete.

Rational Preferences

Definition (Transitive Preference)

Binary relation is called transitive if for every $x, y, z \in X$, whenever xBy and yBz , it must be also true that xBz .

Definition (Rational Preference)

Binary relation $\succsim$ is rational if it is complete and transitive.

- Blood donor relation d is obviously transitive, but not rational, as it is not complete.
- Majority voting relation is always complete, but with the Condorcet Cycle type of population is not transitive, and thus is not rational (I formally define strict preference in a couple of slides):
 - 1 Ann: $x \succ_a y \succ_a z$, Bob: $z \succ_b x \succ_b y$, Charlie: $y \succ_c z \succ_c x$.
 - 2 Majority voting: $x \succ_M y$, $y \succ_M z$, $z \succ_M x$.

Properties of Binary Relations

A few more properties of binary relation B :

- ① **irreflexive**: $\neg xBx$, that is for all $x \in X$ we have $(x, x) \notin B$
- ② **symmetric**: if xBy then yBx , i.e. $(x, y) \in B \Rightarrow (y, x) \in B$
- ③ **asymmetric**: if xBy then $\neg yBx$, i.e. $(x, y) \in B \Rightarrow (y, x) \notin B$
- ④ **antisymmetric**: if xBy and yBx then $x = y$
- ⑤ **negative-transitive**: if $\neg xBy$ and $\neg yBz$ then $\neg xBz$.

Try to prove that if B is asymmetric and negative-transitive, then B is transitive.

Strict Preference

Because of reflexivity, rational preferences capture an *at least as good as* notion. We sometimes use its strict part:

Definition (Strict preference)

Given a rational preference $\succsim$, the strict part (preference) of $\succsim$ is the binary relation $\succ$ defined by:

$$x \succ y \Leftrightarrow x \succsim y \text{ but not } y \succsim x$$

Proposition

Given a rational preference $\succsim$, its strict part $\succ$ is:

- ① always transitive
- ② may or may not be complete
- ③ always irreflexive
- ④ always asymmetric

Indifference

The indifference relation is also very useful (think how often you use indifference curves!)

Definition (Indifference)

Given a rational preference $\succsim$, the indifference part of $\succsim$ is the binary relation $\sim$ defined by:

$$x \sim y \text{ whenever } x \succsim y \text{ and } y \succsim x$$

Proposition

Given a rational preference $\succsim$, its indifference part $\sim$ is:

- ① reflexive
- ② symmetric

Meaning of Rational Preferences

What is the content of the hypothesis that individual preference orders are rational (complete and transitive) binary relations?

Complete and transitive preference relation disallows the agent saying

- “x and y are incomparable” or “I refuse to compare x and y” (e.g., parent refusing to rank his or her children)
- “depends on circumstances” (e.g., on friends’ opinion), or “depends on how it is framed” (e.g., “2 of 3 live” $\succ$ “1 of 3 die”)
- “ $x \sim y$, $y \sim z$ but $x \succ z$ ” (cumulative imperceptible differences)

Individuals subject to thinking costs (due to stress, choice overload) may violate rationality; failure of transitivity may also arise from aggregation of considerations or the use of similarities.

1 Preferences

2 **Utility**

3 Choice

4 Bundles and Time

Identifying the Best Element

Consider the rational preference $\succsim$ over X , and suppose that we want to identify the best element, according to $\succsim$, within a subset of objects $A \subseteq X$. How would you do it? Here is an intuitive algorithm. Select any alternative $x \in A = A_0$. Identify all alternatives that are strictly better than x , A_1 . Repeat

If the number of alternatives is finite, that is enough. Indeed, all subsets have at least one best element

In infinite sets, however, this technique may prove insufficient. Some subsets may fail to have maximal elements and even when they exist, finding them may prove to be difficult

In general, economists prefer easy-to-use optimisation and optimisation techniques based upon utility functions and, often, continuity/differentiability properties.

Utility Representation

Definition (Utility Representation)

Given the rational preference $\succsim$ over X , we say that the real-valued function $u : X \rightarrow \mathbb{R}$ is a **utility representation or function** whenever, for every $x, y \in X$:

$$x \succsim y \Leftrightarrow u(x) \geq u(y)$$

Theorem

Let B be a binary relation on X , and $u : X \rightarrow \mathbb{R}$ such that, for every $x, y \in X$, $xBy \Leftrightarrow u(x) \geq u(y)$. Then, B must be a rational preference.

Proof.

B must be complete because, for any pair x, y , it is either $u(x) \geq u(y)$ meaning xBy or $u(y) \geq u(x)$ meaning yBx or both. B must be transitive because if xBy and yBz , it must be $u(x) \geq u(y)$ and $u(y) \geq u(z)$. Then, $u(x) \geq u(z)$ and, hence, xBz , as desired. $\square$

Preferences on Countable Sets

Theorem

If X is either finite, or at most countably infinite, then any rational preference $\succsim$ on X can be represented by utility function u with a bounded range (e.g., u mapping X to numbers between 0 and 1)

That is, if X does not have “too many elements” then any complete and transitive $\succsim$ on it has a utility representation.

What does it mean that “ X is countably infinite”?

- there is a one-to-one mapping between elements of X and (say) all positive integers (i.e., you can “count” the elements of X)
- for instance, all n -dimensional vectors with integer coordinates form a countable set
- but an interval of reals (e.g., the unit interval) is not countable

Countable and Uncountable Sets

Set $\mathbb{N} = \mathbb{Z}_{++} = \{1, 2, 3, \dots\}$ is countable literally

Set of integers $\mathbb{Z} = \{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\}$ is countable; enumerate elements as:

$$\{0, 1, -1, 2, -2, 3, -3, \dots\}$$

Set of rational numbers $\mathbb{Q} = \{a = p/q, p \in \mathbb{Z}, q \in \mathbb{Z}_{++}\}$ is countable: for $p \neq q$ call $|p| + q$ as number “height” and arrange the numbers by height and sign:

$$\{0/1, 1/1, -1/1, 1/2, -1/2, 2/1, -2/1, \dots\}$$

Interval $[0, 1]$ is not countable:

- if it was, I can enumerate its elements: $[0, 1] = \{x_1, x_2, \dots\}$
- cover each x_n by some interval of length $l_n = 1/10^n$
- then we have

$$1 \leq \sum_{n=1}^{\infty} \frac{1}{10^n} = 1/9$$

a contradiction.

Proof of the Theorem

Suppose that $X = \{x_1, x_2, \dots\}$.

Define two fictitious elements x_{-1} and x_0 and extend preferences by imposing that

$$\forall x_i \in X : x_i \succ x_{-1} \text{ and } x_0 \succ x_i$$

and define $u(x_{-1}) = 0$ and $u(x_0) = 1$.

Take elements in $X = \{x_1, x_2, \dots\}$ in their order.

Define $u(x_k)$ equal to the average of utilities of “next weakly better” and “next weakly worse” elements that have been already assigned utility:

$$u(x_k) = \frac{\max \{u(x_j) | x_k \preceq x_j, j < k\} + \min \{u(x_j) | x_j \preceq x_k, j < k\}}{2}$$

Thus, $u(x_1) = 1/2$ and

- if $x_2 \succ x_1$ then $u(x_2) = 3/4$
- if $x_1 \succ x_2$ then $u(x_2) = 1/4$
- if $x_2 \sim x_1$ then $u(x_2) = 1/2$

Uncountable Choice Sets: Preference Continuity

Most of choice sets you have seen in Core Micro are uncountable, e.g.

$$\{(x, y) \in \mathbb{R}_+ : p_x x + p_y y \leq m\}$$

How do we deal with them?

Intuitively we were working with the indifference curves, or for all $\mathbf{x} \in X \subseteq \mathbb{R}_+^n$ the set of alternatives that are weakly preferred to $\mathbf{x}$ according to preference order.

Standard indifference curves are closed sets: a set $S \subset \mathbb{R}^n$ is closed whenever sequence $\{\mathbf{s}_n\} \rightarrow \mathbf{s}^*$ then $\mathbf{s}^* \in S$. Geometrically, the boundary must belong to the set.

It turns out that in order to generate standard indifference curves we need preference continuity.

Definition (Preference Continuity)

Binary relation $\succsim$ on $X \subseteq \mathbb{R}^n$ is continuous if, for any sequence $\mathbf{x}_1, \mathbf{x}_2, \dots \in X$ which converges to $\mathbf{x}^*$ and $\mathbf{x}_k \succsim \mathbf{y}$ for all $k = 1, 2, \dots$ we also have $\mathbf{x}^* \succsim \mathbf{y}$.

Debreu Theorem

Theorem (Debreu (1954))

Any rational and continuous preference $\succsim$ on $\mathbb{R}_+^n$ can be represented by a continuous utility function $u : \mathbb{R}_+^n \rightarrow \mathbb{R}$.

Proof. To simplify the proof, in addition to assuming that $\succsim$ is rational and continuous on $\mathbb{R}_+^n$ suppose that any consumption bundle $(x_1, \dots, x_n)$ has a **uniform consumption equivalent** $(\hat{x}, \dots, \hat{x})$, i.e. $(x_1, \dots, x_n) \sim (\hat{x}, \dots, \hat{x})$.

Now, for each choice define $u(x_1, \dots, x_n) = \hat{x}$, such that $(x_1, \dots, x_n) \sim (\hat{x}, \dots, \hat{x})$.

- That is, utility is a coordinate where the indifference curve passing through $(x_1, \dots, x_n)$ crosses the “45-degree line”.

Now suppose that $(x_1, \dots, x_n) \sim (\hat{x}, \dots, \hat{x})$ and $(y_1, \dots, y_n) \sim (\hat{y}, \dots, \hat{y})$, then

$$(x_1, \dots, x_n) \succsim (y_1, \dots, y_n) \Leftrightarrow (\hat{x}, \dots, \hat{x}) \succsim (\hat{y}, \dots, \hat{y}) \Leftrightarrow \hat{x} \geq \hat{y} \Leftrightarrow u(x_1, \dots, x_n) \geq u(y_1, \dots, y_n)$$

and therefore u represents $\succsim$.

Lexicographic Preferences

Definition (Lexicographic Preferences)

$(x_1, x_2) L (y_1, y_2)$ if either $x_1 > y_1$ or $x_1 = y_1$ and $x_2 \geq y_2$.

L is complete on $X = \mathbb{R}_+^2$ and is transitive, hence rational. Is L representable?

L is rational, but not continuous:

- Take a sequence $\mathbf{x}_n = (1/n, 0)$ and let $\mathbf{y} = (0, 1)$
- $\mathbf{x}_n L \mathbf{y}$
- $\mathbf{x}_n \rightarrow \mathbf{x}^* = (0, 0)$
- $\mathbf{y} L \mathbf{x}^*$

Lexicographic Preferences

Proposition

Lexicographic preferences over $\mathbb{R}^2$ cannot be represented.

Proof.

- Suppose that L is representable.
- Fix some $x_1 \in [0, 1]$, for each x_1 it must be the case that $u(x_1, 1) > u(x_1, 0)$.
- Then there exist $q(x_1) \in \mathbb{Q} : u(x_1, 1) > q(x_1) > u(x_1, 0)$.
- Such $q(x_1)$ must exist for all $x_1 \in [0, 1]$.
- For each $x_1 \neq x_2$ we have $[u(x_1, 0), u(x_1, 1)] \cap [u(x_2, 0), u(x_2, 1)] = \emptyset$, thus $q(x_1) \neq q(x_2)$.
- However, $[0, 1]$ is uncountable, while set of all q is countable, a contradiction.



Intuition: every point in $\mathbb{R}_+^2$ is its own indifference curve, they cannot cross the 45-degree line.

Monotonic Transformations

Representations of a rational preference $\succsim$ are not unique. The preference is ordinal and, if u represents this preference, any monotone transformation of u represents the same preference

Theorem

Let $\succsim$ be a rational preference over X . Suppose that u represents $\succsim$ and that $f : \mathbb{R} \rightarrow \mathbb{R}$ is strictly increasing. Then, the real-valued function $v = f \circ u$ (that is, $v(x) = f(u(x))$) is also a utility function representing $\succsim$

Proof.

- $x \succsim y$ if and only if $u(x) \geq u(y)$ (because u represents the preferences)
- $u(x) \geq u(y)$ if and only if $v(x) = f(u(x)) \geq f(u(y)) = v(y)$ (because f is strictly increasing)
- thus $x \succsim y$ if and only if $v(x) \geq v(y)$, hence $v = f \circ u$ represents the same preferences as u



Monotonic Transformations: Discussion

Any strictly monotonic transformation of a utility function represents the same preferences because preferences are ordinal (categorical):

- only equivalence classes matter, utility levels do not

Example:

- suppose that $(x_1, x_2) \succsim (y_1, y_2)$ iff $x_1 x_2 \geq y_1 y_2$
- note that $x_1 x_2 \geq y_1 y_2 \Leftrightarrow \ln x_1 + \ln x_2 \geq \ln y_1 + \ln y_2$
- therefore u and $v = \ln u$ represent the same preferences—Cobb-Douglas

Maximisation: Finite Choice Set

Given a rational preference, under what conditions can we guarantee the existence of (at least) one element that is better than all the others?

Proposition

Let $\succsim$ be a rational preference over a finite set X . Every subset $A \subseteq X$ has maximal element.

Proof.

Use pairwise comparisons to find the best element in finitely many steps. □

Maximisation: Infinite Choice Set

With infinite choice set the existence of utility function is not a sufficient condition for well-behave maximisation problem:

- let $u(x) = x$
- let $X = (0, 1)$

Theorem

If $\succsim$ is rational and continuous preference relation over a compact set $X \subseteq \mathbb{R}_+^n$, then X has a maximal element

Definition

A set X is compact if it is bounded and closed.

Proof.

- u representing $\succsim$ is continuous on X (we didn't prove it, but it's true)
- Weierstrass Extreme Value Theorem states that a continuous function on a compact set attains its minimum and maximum.



1 Preferences

2 Utility

3 Choice

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Choice

We have conjectured a model about the mind of the agent, but was it a good conjecture? We can model agent's decisions using choice data:

- we see data on choice, and almost never see data on preferences;
- we want to infer preferences from choice data.

Definition (Choice problem (menu))

A **choice problem (menu)** is a subset of alternatives $A \subseteq X$.

Definition (Choice Data)

Let $\Delta = \{A, B, \dots\}$ be a collection of menus. **Choice Data** is defined as a map of the form

$$c : \Delta \rightarrow X$$

with $c(K) \in K$ for all $K \in \Delta$.

Choice data is given by Δ and $c(\cdot)$:

- Δ is determined by the analyst or occurs naturally in the field;
- c is determined by the agent.

Examples of Choice Data

Universal Choice Data

Universal Choice Data is such that $\Delta = \mathcal{X}$, where $\mathcal{X}$ is the collection of all subsets of X

Binary Choice Data

Binary Choice Data is such that $\Delta = \mathcal{B}$, where $\mathcal{B}$ is the collection of all choice problems formed by two alternatives

Bundle Consumption Choice Data

Let $X = \mathbb{R}_+^n$. Bundle Consumption Choice Data is such that Δ is formed by some budget sets of the form $p_1x_1 + \cdots + p_nx_n \leq m$. Typically, the variation of prices and income may be given by the environment (think of scanner data)

Working Towards Revealed Preference

Consider a menu A . Suppose that we observe $x = c(A)$ and $y \in A$. What can we deduce?

Formally, nothing, even if the individual is rational. It may be that the individual is indifferent among all options

We simplify the analysis and assume that $\succsim$ contains no indifferences (other than reflexivity).

Definition

A rational choice preference relation of finite choice set $c(A, \succ)$ is the unique maximal element of A according to $\succ$.

The notions and techniques can be extended to the general case using set-valued c .

Revealed Preference

Definition (Direct Revelation)

Given choice data c , we say that x is **Directly Revealed Preferred** to $y \neq x$, and write $x \succ_c y$ whenever $x = c(A)$ and $y \in A$

- We have observed $x = c(A)$, with $y \in A$
- We have learnt from this data, and deduce that x must be better than y , $x \succ_c y$ (given our basic assumption)
- Consider now an alternative choice problem A' , with $x, y \in A'$. We may predict that $y \neq c(A')$, because x is a better available option.

Theorem

Suppose that X is finite. Then Universal Choice Data or Binary Choice Data are enough to learn all the preferences of the individual

Proof.

Trivial, because in both cases we observe every binary menu $\{x, y\}$ and can directly learn from this whether $x \succ y$ or $y \succ x$. □

Revealed Preference: Example

Example: College Hall

A college offers repeatedly the same four meals in hall, $\{w, x, y, z\}$. Depending on the time of arrival to hall some options may be unavailable

- On Monday, our student chose x from $\{x, z\}$
- On Tuesday, our student chose w from $\{w, x, y\}$
- On Wednesday, our student choose x from $\{x, y, z\}$

What is the directly revealed preference?

- On Monday, our student chose x from $\{x, z\}$, hence $x \succ_c z$
- On Tuesday, our student chose w from $\{w, x, y\}$, hence $w \succ_c x$ and $w \succ_c y$
- On Wednesday, our student choose x from $\{x, y, z\}$, hence $x \succ_c y$ and $x \succ_c z$
- We cannot compare y and z
- If we impose transitivity, then

Indirect Revealed Preference

Example: College Hall

A college offers repeatedly the same four meals in hall, $\{w, x, y, z\}$. Depending on the time of arrival to hall some options may be unavailable

- On Monday, our student chose x from $\{x, z\}$
- On Tuesday, our student chose w from $\{w, x, y\}$
- On Wednesday, our student choose x from $\{x, y, z\}$

From Tuesday we got that $w \succ_c x$ and from Monday we got that $x \succ_c z$. Does this mean that $w \succ_c z$? Yes, if we believe that the preference is transitive.

Definition (Indirect Revealed Preference)

Given choice data c , we say that x is **Indirectly Revealed Preferred** to $y \neq x$ whenever there is a sequence menus $A_1, \dots, A_n$ and a sequence of choices $z_1 = x, z_2, \dots, z_n = y$ such that $z_i = c(A_i)$ and $z_{i+1} \in A_i$, $i = 1, \dots, n-1$, i.e., whenever $z_1 = x \succ_c z_2 \succ_c \dots \succ_c y = z_n$.

- $A_1 = \{w, x, y\}$, $c(A_1) = w$ and $x \in A_1$.
- $A_2 = \{x, z\}$, $c(A_2) = x$ and $z \in A_2$.
- Hence $w \succ_c x \succ_c z$.

Learning Preferences

What is the minimum number of choice problems that we need to observe to learn all the preferences of the individual? Which is the needed structure?

If X contains n elements, $n - 1$ choice problems may be enough to learn all the preferences:

- ① **Confirming what we know.** If we conjecture that $x_1 \succ x_2 \succ \cdots \succ x_n$ is the preference, we just need to observe $n - 1$ binary problems, $\{x_i, x_{i+1}\}$, to confirm what we conjecture
- ② **Learning in the lab.** We can run $n - 1$ experiments sequentially. Ask for X . The chosen element x_1 must be the best. Ask for $X \setminus \{x_1\}$. The chosen element x_2 must be the second best. And so on.

Rational Choice Data

Most of our models start by assuming that individuals have rational preferences $\succsim$.

Actually, when we defined the indirect revealed preference relationship, we imposed transitivity.

As a (social) science, we want to also test the theory. Having observed c , can we:

- reject the hypothesis that the individual has rational preferences?
- write down the type of conditions that we need to check for that test?

Definition (Rational Choice Data I)

Choice Data $c : \Delta \rightarrow X$ is said to be **rationalizable** if there exists a rational preference $\succsim$ such that $c(A)$ is always the maximal element in A according to $\succsim$.

Definition (Rational Choice Data II)

Choice Data $c : \Delta \rightarrow X$ is said to be **rationalizable** if there exists a utility representation u such that $c(A)$ is always the maximal element in A according to u .

Property α

Suppose that the student is in the queue and, having seen what is available today, say $A = \{x, y, z\}$, decides to go for meal x . The student in front takes the last portion of meal y , leaving effectively menu $B = \{x, z\}$ for our student. What should be the choice now?

A rational student would simply choose x .

Definition (Property α , Sen's α)

We say that c satisfies property α if for any pair of choice problems $B \subseteq A$, we have

$$c(A) \in B \Rightarrow c(B) = c(A)$$

Property α and Rationalizability

Theorem

If choice data is rationalizable, it must satisfy property α

Proof.

- If c is rationalizable, there exists a preference $\succsim$ explaining all choices.
- That is, for any $x \in c(A)$ and $y \in A$ we have $x \succsim y$.
- Then, for each $y \in B \subseteq A$ we have $x \succsim y$.



Note, that property α implies rationalizable preferences only for strict domain. In general property β is necessary:

Definition (Property β)

We say that c satisfies property α if for any pair of choice problems $B \subseteq A$, we if $x, y \in c(B)$ and $x \in C(A)$ then $y \in C(A)$.

Rational Choice Data: Example

Example: College Hall Revisited

We observe two students at Hall over an entire year. Given the variation in arrival time, we got universal choice data from them. It is as follows

Menu	Ann	Bob
$\{M, F, V\}$	V	F
$\{M, F\}$	F	M
$\{M, V\}$	V	V
$\{F, V\}$	V	V

- Ann's choice data are clearly rational. The preference $V \succ_A F \succ_A M$ explains all choices
- Bob's choice data are clearly not rational. No preference can explain all choices:
 - Any preference in which $M \succ_B F$ fails to explain the choice from menu $\{M, F, V\}$.
 - Any preference in which $F \succ_B M$ fails to explain the choice from menu $\{M, F\}$.

Rational Choice Data on the Universal Domain

That is all we need with universal choice data

Theorem

Let X finite. If the universal choice data c satisfies property α , c is rationalizable

Proof.

We construct the preference as follows:

- take $A_1 = X$ and $x_1 = c(A_1)$
- by property α x_1 is chosen whenever $x_1 \in A \subseteq X$
- let $A_2 = A_1 \setminus \{x_1\}$ and $x_2 = c(A_2)$
- by property α x_2 is chosen whenever $x_2 \in A$ and $x_1 \notin A$
- define $x_1 \succ_c x_2$
- let $A_3 = A_2 \setminus \{x_2\}$ and $x_3 = c(A_3)$
- ...
- in finite number of steps reach $x_1 \succ_c x_2 \succ_c \dots \succ_c x_K$ where K is the number of elements in X .



WARP

Here is an alternative condition for rational choice.

Definition

Weak Axiom of Revealed Preference Let $x, y \in A$ and $x, y \in B$, then we cannot have $c(A) = x$ and $C(B) = y$

Remark

Note, that this is not exactly the same as alpha: here we have $\{x, y\} \in A \cap B$, but not necessarily $B \subseteq A$. So, it may be not true that $C(B) = x$, but it must be true that $C(B) \neq y$.

Theorem

Choice induced by rational preference $\succsim$ satisfies WARP; if X is finite and c satisfies WARP on universal choice data, then there exists a rational preference relation that induces c .

4. Rational Choice Data

Example: College Hall Revisited

Menu	Ann	Bob
$\{M, F, V\}$	V	F
$\{M, F\}$	F	M
$\{M, V\}$	V	V
$\{F, V\}$	V	V

- It is easy to check that Ann satisfies the announced properties.
- You can see that Bob fails the properties by checking $\{M, F, V\}$ and $\{M, F\}$:
 - The choice in the largest menu differs from the choice in the smallest menu, so α fails.
 - Equivalently, F is revealed preferred to M in the largest menu, but M is revealed preferred to F the smallest menu, i.e., the weak axiom of revealed preference fails.

Rational Choice?!

Second-largest piece of cake

A gluttonous kid has been told to be polite and avoid taking the largest piece. The kid always takes the second-largest piece of cake

- $c(\{1, 2, 3\}) = 2$
- $c(\{1, 2\}) = 2$
- $c(\{1, 3\}) = 3$
- $c(\{2, 3\}) = 3$

Do this choice data violate property α and/or WARP?

1 Preferences

2 Utility

3 Choice

4 Bundles and Time

Bundles and Time

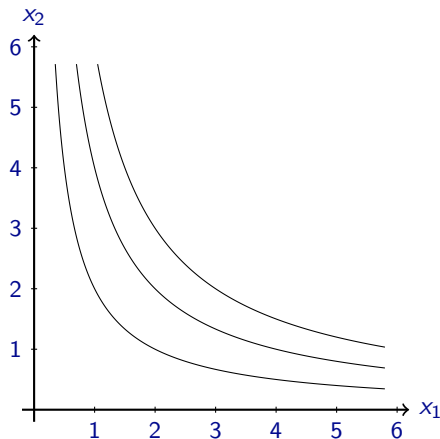
Up to now, we have mostly contemplated an abstract and flexible model of behaviour. Here are some slightly more applied questions that economists have in mind:

- How do people take into consideration the multi-dimensional nature of their decisions, e.g., when consumption of multiple goods is involved?
- How do people take into consideration the inherent temporal dimension of their decisions?
- How do people take into consideration the inherent risk dimension of their decisions?

We now briefly expand on the first two questions. Week 6 will be completely devoted to the the third one.

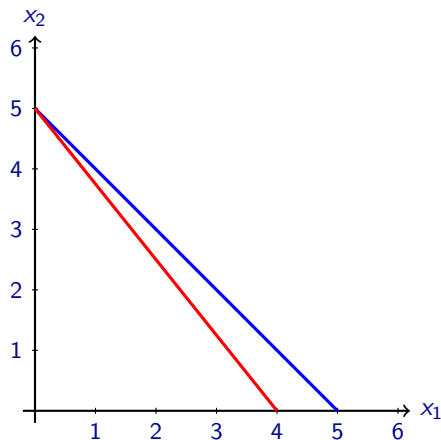
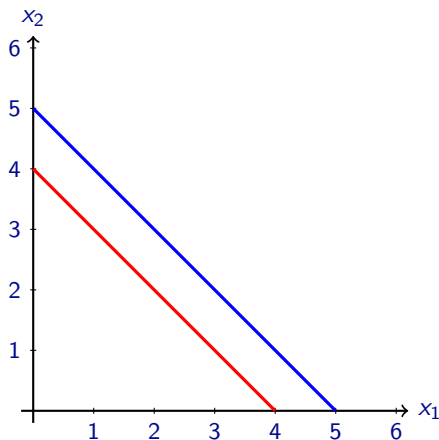
Indifference Curves

The traditional model on the consumption of different goods. The space is $X = \mathbb{R}_+^n$. Most examples use $n = 2$

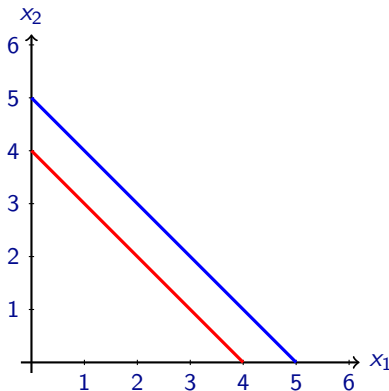


Income and Price Variations

In this framework, universal menus are not reasonable. Variation of menus is usually the result of variation of **income** or **prices** or both.



Monotone Preferences



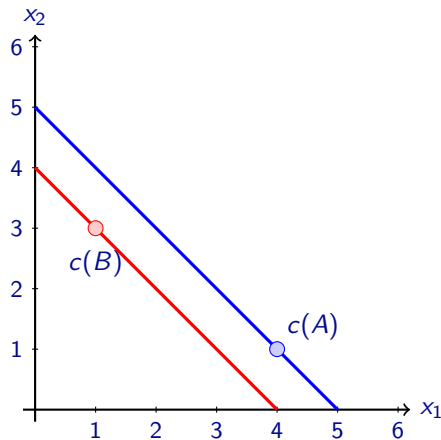
- For general preferences there is no natural 'order' in the interior of the budget set.
- Continuity of preferences gives nice equivalence classes—indifference curves, but puts no restrictions on the shape or direction of utility change.
- Once extra condition of monotonicity is imposed, price-quantity data allows us to say more about the preferences.

Definition

Strict Monotonicity A preference relation $\succ$ on $X \subseteq \mathbb{R}_+^n$ is strictly monotone if for all $\mathbf{x}, \mathbf{y} \in X$ such that $\mathbf{x} > \mathbf{y}$ we have $\mathbf{x} \succ \mathbf{y}$.

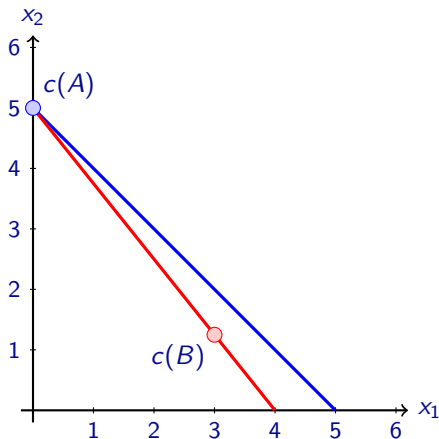
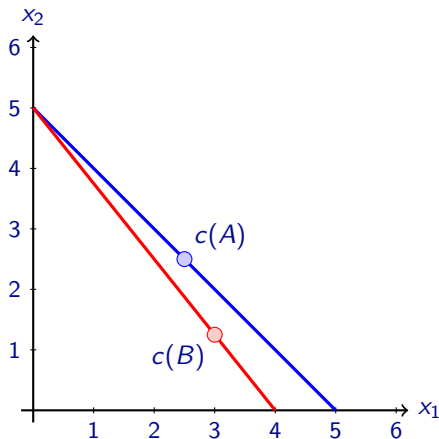
Recall that $\mathbf{x} > \mathbf{y}$ means $x_i \geq y_i$ for all $i = 1, \dots, n$ and there exists j such that $x_j > y_j$.

Change in Income



Notice how the choice of the big (budget) set is never available in the small (budget) set. Property α or the weak axiom of revealed preference say nothing

Change in Price

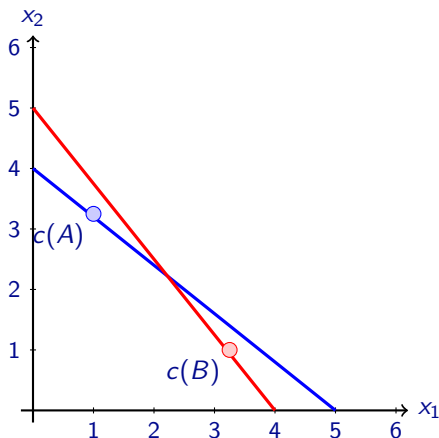
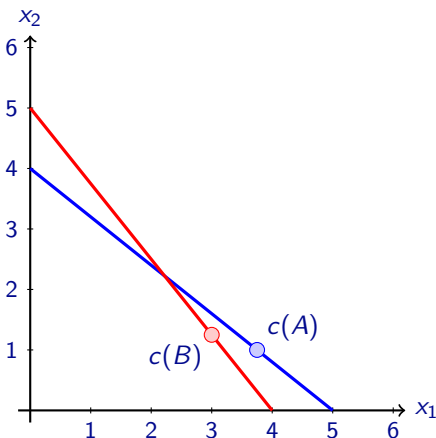


Notice how the choice of the big (budget) set is not available in the small (budget) set.

Property α cannot be rejected, except in the very extreme case of the right graph

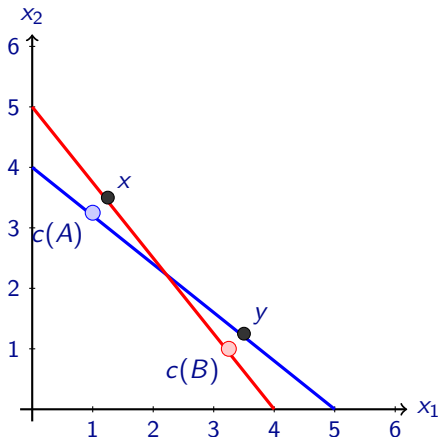
Change in both Price and Income

Combined variations are more informative. Here we use the weak axiom of revealed preference jointly with monotonicity, even accepting that indifferences may exist.



Which one do you think may represent rational preferences?

Change in both Price and Income



- Due to strict monotonicity $x \succ c(A)$.
- By revealed preference $c(B) \succsim x$
- By revealed preference $c(A) \succsim c(B)$
- Thus,

$$x \succ c(A) \succsim c(B) \succsim x$$

a contradiction.

This is an example of choice data violating the WARP.

Definition (WARP for Bundles)

Suppose x_1 was chosen given prices p_1 , x_2 was chosen given prices p_2 and $p_1 x_2 \leq p_1 x_1$. Then it cannot be the case that $p_2 x_2 \geq p_2 x_1$.

Other Useful Properties

Definition ((strict) Convexity)

Preferences are strictly convex if for every $x, y \in X \subseteq \mathbb{R}_+^n$ such that $x \sim y$ and $x \neq y$ and every $\alpha \in (0, 1)$:

$$\alpha x + (1 - \alpha)y \succ x \sim y$$

This is a property required for the uniqueness of the demand.

Consider vector (x, m) describing the amount of some good and the amount of money. Remember that quasi-linear preferences allow for the representation $U(x, m) = u(x) + m$

Definition (Quasi-linearity)

The preference is quasi-linear on $(x, m) \in \mathbb{R}_+^{n+1}$ if for all x, y, m, m', n we have

$$(x, m) \succeq (y, m') \Leftrightarrow (x, m + n) \succeq (y, m' + n)$$

Time and Discounted Utility

The traditional model to deal with time requires us to think of prizes and time at the same time. We can nicely use the simplest set $X = \mathbb{R}_+^2$ to illustrate it

An element (x, t) describes the amount of money x to be paid at time t .

The standard model in economics is that of discounted utility. It involves an (increasing) utility over money, $u : \mathbb{R}_+ \rightarrow \mathbb{R}$ and a discount factor $\delta \in (0, 1]$

Definition (Discounted Utility)

$$(x, t) \succsim (y, s) \Leftrightarrow U(x, t) = \delta^t u(x) \geq \delta^s u(y) = U(y, s)$$

Stationarity

Discounted Utility is a rational preference. It can be represented by a utility function (function U). Naturally, we can use property α or the weak axiom of revealed preference to test such rationality when evaluating choices of dated prizes (x, t)

Importantly, discounted utility has an extra consistency property that can be formalised and tested:

Definition (Stationarity)

We say that the preference $\succsim$ over dated prizes satisfies stationarity if for every x, y, t, s, r :

$$(x, t) \succsim (y, s) \Leftrightarrow (x, t + r) \succsim (y, s + r)$$

Time Preferences Properties

Fishburn and Rubinstein (1982) show that discounted utility involves

- Rational Preferences
- Monotone Preferences (more money is better)
- Some continuity property
- Stationarity

Less Useful Properties

Definition (Single-Peakedness)

Let $X = [0, 1]$. We say that $\succsim$ is single-peaked if there exists x^* such that:

- $y_2 < y_1 < x^* \Rightarrow y_1 \succ y_2$
- $x^* < y_1 < y_2 \Rightarrow y_1 \succ y_2$

Think how to generalise this definition to $\mathbb{R}^n$.

Definition (Local Non-Satiation)

Let $x \in \mathbb{R}_+^n$. Preferences $\succsim$ called locally non-satiated, if for any $x \in X$ there exists $\varepsilon > 0$ and $y \in X$ such that

$$\|y - x\| < \varepsilon \quad \text{and} \quad y \succ x$$

Can you construct locally non-satiated preferences that are not monotone?